SquarePoint

Campus: IIT Delhi · Role: Software Developer (SDE)

1. IIT Delhi — Software Developer (SDE)

Q1. Minimum Time to Finish All Instructions

There are n servers and m instructions. You are given an array arr of length m, where $arr[i]$ is the server the i-th instruction is assigned to.

Every instruction must be executed completely by exactly one server. All servers start at time 0 and work in parallel, and each server executes its instructions one after another, without interruption.

RULES

- If an instruction is executed by the server it is **assigned to**, it takes 1 second.
- If it is executed by **any other** server, it takes 2 seconds.

Return the **minimum time** required to complete all the instructions.

> Note (approach used — passed the visible test cases; hidden tests were not shown): binary search on the answer x. For a candidate x, each server first executes up to x of its own instructions; let t_i be its spare time left. Server i can then take $\lfloor t_i/2 \rfloor$ other instructions, so x is feasible if $\sum_i \lfloor t_i/2 \rfloor$ is at least the number of instructions still left over. Feasibility is monotonic in x.

> Practice (closest): Codeforces 1701C — Schedule Management

Q2. Text Editor

Implement a simple text editor. At any moment its state is the **current text** s and a **cursor position** c, with $0 \le c \le |s|$ (the cursor sits between characters). Initially the text is empty and $c = 0$.

Process the operations in the given order:

- `left(x)` — move the cursor x positions to the left. It cannot go past the beginning of the text.
- `right(x)` — move the cursor x positions to the right. It cannot go past the end of the text.
- `insert(val)` — insert the string `val` at the cursor position; the cursor moves to the end of the inserted text.
- `delete(x)` — delete x characters to the **right** of the cursor (fewer if the text ends earlier). The cursor does not move.
- `backspace(x)` — delete x characters to the **left** of the cursor (fewer if the text starts earlier). The cursor moves left by the number of characters deleted.
- `print()` — record the current text.

Return the results of all `print()` operations, in order, as a `vector<string>`.

> Practice (closest): LeetCode 2296 — Design a Text Editor